

$$(a) \quad |1,1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{cases} \hat{S}_+ |1,1\rangle = 0 \\ \hat{S}_+ |1,0\rangle = \sqrt{2}\hbar |1,1\rangle \\ \hat{S}_+ |1,-1\rangle = \sqrt{2}\hbar |1,0\rangle \end{cases}$$

$$\begin{cases} \hat{S}_- |1,1\rangle = \sqrt{2}\hbar |1,0\rangle \\ \hat{S}_- |1,0\rangle = \sqrt{2}\hbar |1,-1\rangle \\ \hat{S}_- |1,-1\rangle = 0 \end{cases}$$

$$\begin{cases} \hat{S}_z |1,1\rangle = \hbar |1,1\rangle \\ \hat{S}_z |1,0\rangle = 0 \\ \hat{S}_z |1,-1\rangle = -\hbar |1,-1\rangle \end{cases}$$

$$\Rightarrow \hat{S}_+ = \sqrt{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \hat{S}_- = \sqrt{2}\hbar \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad \hat{S}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-) = \frac{\sqrt{2}}{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{S}_y = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-) = \frac{\sqrt{2}}{2}i\hbar \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

(b) $\hat{S}_y$ 的本征值与本征态为:

$$|1,1\rangle_y = \frac{1}{2} \begin{pmatrix} \sqrt{2}i \\ 1 \\ -1 \end{pmatrix}, \quad S_y = \hbar$$

$$|1,0\rangle_y = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \quad S_y = 0$$

$$|1,-1\rangle_y = \frac{1}{2} \begin{pmatrix} -\sqrt{2}i \\ 1 \\ 1 \end{pmatrix}, \quad S_y = -\hbar$$

$$\text{则分别计算概率: } |\langle 1,1|_y |1,0\rangle|^2 = \left| \frac{1}{2}(1, -\sqrt{2}i, -1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2}$$

$$|\langle 1,0|_y |1,0\rangle|^2 = \left| \frac{\sqrt{2}}{2}(1, 0, 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right|^2 = 0$$

$$|\langle 1,-1|_y |1,0\rangle|^2 = \left| \frac{1}{2}(1, \sqrt{2}i, 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2}$$

则 50% $\hbar$, 50% $-\hbar$. (也可由对称性一眼看出答案)

$$(c) \text{ 初态为 } |1,1\rangle_y = \frac{1}{2} \begin{pmatrix} \sqrt{2}i \\ 1 \\ -1 \end{pmatrix}$$

$$\text{由 } H = -\gamma \vec{B} \cdot \vec{S} \quad \text{时间演化因子分别为 } m_s = +1: e^{-iE_+ t/\hbar} = e^{i\gamma B_0 t}$$

$$m_s = 0: 1$$

$$m_s = -1: e^{-i\gamma B_0 t}$$

$$\text{则 } |\psi(t)\rangle = \frac{1}{2} \begin{pmatrix} e^{i\gamma B_0 t} \\ \sqrt{2}i \\ -e^{-i\gamma B_0 t} \end{pmatrix}$$

$$\langle \hat{S}_y \rangle = \langle \psi(t) | \hat{S}_y | \psi(t) \rangle$$

$$= \frac{1}{2} (e^{-i\gamma B_0 t}, -\sqrt{2}i, -e^{i\gamma B_0 t}) \cdot \frac{i\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} e^{i\gamma B_0 t} \\ \sqrt{2}i \\ -e^{-i\gamma B_0 t} \end{pmatrix}$$

$$= \frac{i\hbar}{4\sqrt{2}} (e^{-i\gamma B_0 t}, -\sqrt{2}i, -e^{i\gamma B_0 t}) \cdot \begin{pmatrix} -\sqrt{2}i \\ 2\cos(\gamma B_0 t) \\ \sqrt{2}i \end{pmatrix}$$

$$= \frac{\hbar}{4} (e^{-i\gamma B_0 t}, 1, -e^{i\gamma B_0 t}) \begin{pmatrix} 1 \\ 2\cos(\gamma B_0 t) \\ -1 \end{pmatrix} = \frac{\hbar}{4} \cdot 4\cos(\gamma B_0 t) = \hbar \cdot \cos(\gamma B_0 t).$$



(a) 谐振子的本征态为 $|n\rangle$, $E_n = (n + \frac{1}{2})\hbar\omega$, $n=0, 1, 2, \dots$

对于玻色子, 基态 $n_1 = n_2 = 0$, 第一激发态 $n_1, n_2 = 0, 1$:

$$|\psi_{\text{ground}}\rangle = |0\rangle \otimes |0\rangle$$

$$|\psi_{1st}\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle)$$

(b) 对于费米子, 基态 $n_1, n_2 = 0, 1$, 第一激发态 $n_1, n_2 = 0, 2$:

$$|\psi_{\text{ground}}\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle) \quad (\text{可多负号})$$

$$|\psi_{1st}\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |2\rangle - |2\rangle \otimes |0\rangle) \quad (\text{可多负号})$$

(c) 费米子总量子态交换反对称.

能量最低态(轨道) $n_1 = n_2 = 0$ 为对称态, 则自旋应为单重态 (singlet)

$$|\psi_{\text{ground}}\rangle = (|0\rangle_1 \otimes |0\rangle_2) \otimes \frac{1}{\sqrt{2}}(|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 - |\downarrow\rangle_1 \otimes |\uparrow\rangle_2)$$

(可不带下标, 可省略 $\otimes$ 符号)

第一激发态 ~~$n_1, n_2 = 0, 1$~~ $n_1, n_2 = 0, 1$ 可为对称态 $\frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle)$

也可为反对称态 ~~$\frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$~~ $\frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$

分别对应反对称的单重态和对称的三重态.

$$\text{则 } |\psi_{1st}\rangle_1 = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle) \otimes \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|\psi_{1st}\rangle_{2,3,4} = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle) \otimes \begin{cases} |\uparrow\rangle \otimes |\uparrow\rangle \\ \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \\ |\downarrow\rangle \otimes |\downarrow\rangle \end{cases}$$

(d) 设 $\hat{S}_1 + \hat{S}_2 = \hat{S}$ (总自旋)

$$\text{则 } \hat{S}^2 = (\hat{S}_1 + \hat{S}_2)^2 = \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1 \cdot \hat{S}_2$$

$$\Rightarrow \hat{S}_1 \cdot \hat{S}_2 = \frac{1}{2}(\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2)$$

$$= \frac{1}{2}(s(s+1) - s_1(s_1+1) - s_2(s_2+1))\hbar^2$$

$$= \begin{cases} \frac{1}{4}\hbar^2, s=1 \\ -\frac{3}{4}\hbar^2, s=0 \end{cases}$$

基态无简并, 且仅可能为单重态, 所以 $\langle \psi_{\text{ground}} | H' | \psi_{\text{ground}} \rangle = -\frac{3}{4}\hbar^2\beta$, 有一个偏移.

第一激发态有四重简并, 但 ~~$\hat{S}_1, \hat{S}_2$~~ 同时为 $\hat{S}_1, \hat{S}_2$ 的本征态, 所以简并微扰矩阵只有对角元非零.

$$\begin{cases} \Delta E_1 = -\frac{3}{4}\hbar^2\beta & (\text{单重态}) \quad \text{四个态} \\ \Delta E_2 = +\frac{1}{4}\hbar^2\beta & (\text{三重态}) \end{cases}$$

~~第一激发态~~

